

— After Remark —

27/02/19

Wishart Distr:

let $z = (z_{ij})_{p \times p}$

$$\begin{aligned} K(p, n) &= \left[2^{\frac{np}{2}} \pi^{\frac{p(p-1)}{4}} \prod_{i=1}^p \left| \frac{n-i+1}{2} \right| \right]^{-1} \\ &= \left[2^{n/2} \pi^0 \prod_{i=1}^p \left| \frac{n-i+1}{2} \right| \right]^{-1} \\ &= \left[2^{n/2} \prod_{i=1}^p \left| \frac{n-i+1}{2} \right| \right]^{-1} \end{aligned}$$

$$f(z) = K(p, n) |\Sigma|^{-1/2} |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr } \Sigma^{-1} z \right\}$$

$$p=1 \text{ and } \Sigma = \sigma^2$$

$$\begin{aligned} f(z) &= \left[2^{n/2} \prod_{i=1}^p \left| \frac{n-i+1}{2} \right| \right]^{-1} (\sigma^2)^{-1/2} |z|^{n-2} \exp \left\{ -\frac{1}{2} (\sigma^2)^{-1} z \right\} \\ &= \left[2^{n/2} \prod_{i=1}^p \left| \frac{n-i+1}{2} \right| \right]^{-1} \frac{1}{(\sigma^2)^{1/2}} (z)^{\frac{n}{2}-1} \exp \left\{ -\frac{z}{2\sigma^2} \right\} \end{aligned}$$

$$= \frac{1}{2^{n/2} \sqrt{\eta_2}} \frac{1}{(\sigma^2)^{1/2}} (z)^{\frac{n}{2}-1} \exp \left\{ -\frac{z}{2\sigma^2} \right\}$$

$$= \frac{1}{2^{n/2} \sqrt{\eta_2}} \left(\frac{z}{\sigma^2} \right)^{\frac{n}{2}-1} \frac{1}{\sigma^2} \exp \left\{ -\frac{z}{2\sigma^2} \right\}$$

$$\text{let } \frac{z}{\sigma^2} = t \Rightarrow \frac{dz}{\sigma^2} = dt \Rightarrow \frac{dz}{dt} = \sigma^2$$

$$\begin{aligned} f(t) = f(z) |J| &= \frac{1}{2^{n/2} \sqrt{\eta_2}} t^{\frac{n}{2}-1} \frac{1}{\sigma^2} \exp \left\{ -\frac{t}{2} \right\} \times \sigma^2 \\ &= \frac{1}{2^{n/2} \sqrt{\eta_2}} t^{\frac{n}{2}-1} e^{-t/2} \end{aligned}$$

$$\therefore T = \frac{z}{\sigma^2} \sim \chi^2_{n/2}$$

Characteristic fn of Wishart Distribution:

$$\begin{aligned}\phi(t) &= \phi_z(t_{11}, t_{12}, \dots, t_{1p}, t_{21}, \dots, t_{2p}, \dots, t_{pp}) \\ &= E \left[\exp \left[i (t_{11} z_{11} + t_{12} z_{12} + \dots + t_{pp} z_{pp}) \right] \right] \\ &= E \left[\exp \left\{ \frac{i}{2} (2 t_{11} z_{11} + 2 t_{12} z_{12} + \dots + 2 t_{pp} z_{pp}) \right\} \right]\end{aligned}$$

Define, $b_{ij} = (1 + \delta_{ij}) t_{ij}$ where B is a $p \times p$ matrix. $\delta_{ij} = 1$ if $i=j$, $\delta_{ij} = 0$ if $i \neq j$.

$$\begin{aligned}\text{if } i=j, \delta_{ij} &= 1 \\ b_{ij} &= (1+1)t_{ij} \\ b_{ij} &= 2t_{ij} \\ \text{if } i \neq j, \delta_{ij} &= 0 \\ b_{ij} &= (1+0)t_{ij} \\ b_{ij} &= t_{ij}\end{aligned}$$

$$\therefore \phi(t) = E \left\{ \exp \left[\frac{i}{2} (b_{11} z_{11} + b_{22} z_{22} + \dots + b_{pp} z_{pp} + \dots + 2b_{12} z_{12} + 2b_{13} z_{13} + \dots) \right] \right\}$$

$$= E \left[\exp \left\{ \frac{i}{2} (\text{tr } BZ) \right\} \right]$$

$$BZ = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{12} & b_{22} & \dots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1p} & b_{2p} & \dots & b_{pp} \end{bmatrix} \begin{bmatrix} z_{11} & z_{12} & \dots & z_{1p} \\ z_{12} & z_{22} & \dots & z_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ z_{1p} & z_{2p} & \dots & z_{pp} \end{bmatrix}$$

$$BZ = \begin{bmatrix} b_{11} z_{11} + b_{12} z_{12} + \dots + b_{1p} z_{1p} \\ b_{12} z_{12} + b_{22} z_{22} + \dots + b_{2p} z_{2p} \\ \vdots \\ b_{1p} z_{1p} + b_{2p} z_{2p} + \dots + b_{pp} z_{pp} \end{bmatrix}$$

$$\therefore \text{tr}(BZ) = b_{11} z_{11} + b_{22} z_{22} + \dots + b_{pp} z_{pp} + 2b_{12} z_{12} + \dots + 2b_{1p} z_{1p} + \dots$$

$$\therefore \phi(t) = \int \exp \left\{ \frac{i}{2} [\text{tr}(Bz)] \right\} f(z) dz$$

$$= \int \exp \left\{ \frac{i}{2} (\text{tr}(Bz)) \right\} K(p, n) |\Sigma^{-1}|^{n/2} |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr}(z) \right\} dz$$

$$= K(p, n) |\Sigma^{-1}|^{n/2} \int_z |z|^{n-p-1} \exp \left\{ -\frac{1}{2} (\text{tr} \Sigma^{-1} z - i \text{tr}(Bz)) \right\} dz$$

$$= K(p, n) |\Sigma^{-1}|^{n/2} \int_z |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr} (\Sigma^{-1} - iB) z \right\} dz$$

$$= K(p, n) |\Sigma^{-1}|^{n/2} \int_z |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr} \Sigma_0^{-1} z \right\} dz$$

where $\Sigma_0^{-1} = \Sigma^{-1} - iB$

$$= \frac{|\Sigma^{-1}|^{n/2} |\Sigma_0^{-1}|^{n/2}}{|\Sigma_0^{-1}|^{n/2}} K(p, n) \int_z |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr} \Sigma_0^{-1} z \right\} dz$$

$$= \frac{|\Sigma^{-1}|^{n/2}}{|\Sigma_0^{-1}|^{n/2}} \underbrace{\left(K(p, n) |\Sigma_0^{-1}|^{n/2} \int_z |z|^{n-p-1} \exp \left\{ -\frac{1}{2} \text{tr} \Sigma_0^{-1} z \right\} dz \right)}_1$$

$$= \frac{|\Sigma^{-1}|^{n/2}}{|\Sigma_0^{-1}|^{n/2}} = \frac{1}{|\Sigma|^{n/2} |\Sigma_0^{-1}|^{n/2}} = \frac{1}{|\Sigma \Sigma_0^{-1}|^{n/2}}$$

$$= \frac{1}{|\Sigma [\Sigma^{-1} - iB]|^{n/2}}$$

$$= \frac{1}{|I - iB \Sigma|^{n/2}}$$

$$\therefore \phi(t) = \frac{1}{|I - iB \Sigma|^{n/2}}$$

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$$\therefore k = \frac{1}{(2\pi)^{p/2} \sqrt{\sigma_{11}\sigma_{22}(1-\sigma_{12})^2}}$$

12/19 (Xerox)

i) Result:

If $X_k \sim N_p(0, \Sigma)$ $\forall k = 1, 2, \dots, n$, then, $S = \sum_{k=1}^n X_k X_k' \sim N_p(0, \Sigma)$

Proof:

$X_k \sim N_p(0, \Sigma)$

$$f(X_k) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} X_k' \Sigma^{-1} X_k \right\}$$

$$= \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} \text{tr}(\Sigma^{-1} X_k' X_k) \right\}$$

(using result ① on page 12)

$$\phi_S(B) = E \left[\exp \left\{ \frac{i}{2} \text{tr} B S \right\} \right]$$

$$= E \left[\exp \left\{ \frac{i}{2} \text{tr} B \left(\sum_{k=1}^n X_k X_k' \right) \right\} \right]$$

$$= E \left[\prod_{k=1}^n \exp \left\{ \frac{i}{2} \text{tr} (B X_k X_k') \right\} \right]$$

$$= \prod_{k=1}^n E \left\{ \exp \left\{ \frac{i}{2} \text{tr} (B X_k X_k') \right\} \right\}$$

$$= \prod_{k=1}^n \int_{\mathcal{X}_k} \exp \left\{ \frac{i}{2} \text{tr} (B X_k X_k') \right\} f(X_k) dX_k$$

$$= \prod_{k=1}^n \int_{\mathcal{X}_k} \exp \left\{ \frac{i}{2} \text{tr} (B X_k X_k') \right\} \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} \text{tr} (\Sigma^{-1} X_k' X_k) \right\} dX_k$$

$$= \prod_{k=1}^n \int_{x_k} \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} \text{tr} (\bar{\Sigma}^{-1} x_k x_k' - i B x_k x_k') \right\} dx_k$$

$$= \prod_{k=1}^n \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \int_{x_k} \exp \left\{ -\frac{1}{2} \text{tr} (\bar{\Sigma}^{-1} - i B) x_k x_k' \right\} dx_k$$

$$= \prod_{k=1}^n \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \int_{x_k} \exp \left\{ -\frac{1}{2} \text{tr} \bar{\Sigma}_0^{-1} x_k x_k' \right\} dx_k$$

$$\left\{ \begin{array}{l} \text{where } \bar{\Sigma}_0^{-1} = \bar{\Sigma}^{-1} - i B \\ \Sigma_0 = (\bar{\Sigma}^{-1} - i B)^{-1} \end{array} \right\}$$

$$= \prod_{k=1}^n \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} (2\pi)^{p/2} |\Sigma_0|^{1/2}$$

$$= \prod_{k=1}^n \left\{ \frac{1}{|\Sigma|^{1/2}} |\bar{\Sigma}^{-1} - i B|^{1/2} \right\}$$

$$= \prod_{k=1}^n \left\{ \frac{1}{|\Sigma|^{1/2}} \times \frac{1}{|\bar{\Sigma}^{-1} - i B|^{1/2}} \right\}$$

$$= \prod_{k=1}^n \left\{ \frac{1}{|\Sigma (\bar{\Sigma}^{-1} - i B)|^{1/2}} \right\}$$

$$= \prod_{k=1}^n \left\{ \frac{1}{|I - i B \Sigma|^{1/2}} \right\}$$

$$= \frac{1}{|I - i B \Sigma|^{n/2}}$$

$$\therefore S = \sum_{k=1}^n x_k x_k' \sim W_p(n, \Sigma)$$

By uniqueness property of characteristic fn.

Statement - If $Z_1 \sim N_p(n_1, \Sigma)$
 $Z_2 \sim N_p(n_2, \Sigma)$
 then $Z_1 + Z_2 \sim N_p(n_1 + n_2, \Sigma)$

(2) Result: (Additive Property)

If $Z_i \sim N_p(n_i, \Sigma)$ where $i=1, 2$, then, obtain the distn of $Z_1 + Z_2$ where Z_1 and Z_2 are independent.

Proof:

Let $Y = Z_1 + Z_2$

$$\begin{aligned} \phi_Y(t) &= E[e^{it'Y}] \\ &= E[e^{it'(Z_1 + Z_2)}] \\ &= E[e^{it'Z_1} \cdot e^{it'Z_2}] \\ &= E[e^{it'Z_1}] \cdot E[e^{it'Z_2}] \\ &= \frac{1}{|I - iB\Sigma|^{n_1/2}} \cdot \frac{1}{|I - iB\Sigma|^{n_2/2}} \\ &= \frac{1}{|I - iB\Sigma|^{\frac{n_1 + n_2}{2}}} \end{aligned}$$

This is characteristic fn of $N_p(n_1 + n_2, \Sigma)$

$\therefore$ By uniqueness property,

$$Y = Z_1 + Z_2 \sim N_p(n_1 + n_2, \Sigma).$$

(3) Result:

If $Z \sim N_p(n, \Sigma)$ and C is $q \times p$ matrix of constants with $\text{rank} = q$ then $CZC' \sim N_q(n, C\Sigma C')$

Proof:

If $X_k \sim N_p(0, \Sigma)$

where, $k=1, 2, \dots, n$ and are independent then,

$$\sum X_k X_k' \sim N_p(n, \Sigma)$$

Let C be a matrix with rank ' q ' then,

$$CX \sim N_p(0, C\Sigma C')$$

$\therefore X_i$'s are independent, $C X_i$'s are also independent.

Hence,

$$\sum_{k=1}^n C X_k (C X_k)' \sim W_q(n, C \Sigma C')$$

$$\therefore \sum_{k=1}^n C X_k X_k' C' \sim W_q(n, C \Sigma C')$$

$$\therefore C \left(\sum_{k=1}^n X_k X_k' \right) C' \sim W_q(n, C \Sigma C')$$

$$\therefore C Z C' \sim W_q(n, C \Sigma C')$$

$$\text{where } Z = \sum_{k=1}^n X_k X_k'$$

~~Unit = 2~~

let $X_k \sim N(0, \Sigma)$ $k=1, 2, \dots, n$ are indep.

then, $\sum X_k X_k' \sim W_q(n, \Sigma)$

let $C_{q \times p}$ be a matrix of rank $= q$

then,

$$C X_k \sim N(0, C \Sigma C')$$

and,

$$\sum (C X_k) (C X_k)' \sim W_q(n, C \Sigma C')$$

$$\sum C X_k X_k' C' \sim W_q(n, C \Sigma C')$$

$$C \left(\sum X_k X_k' \right) C' \sim W_q(n, C \Sigma C')$$

$$C Z C' \sim W_q(n, C \Sigma C') //$$